

OKC SHOT PROBABILITY MODEL

Predicting shot make probability with interpretable validation and XGBoost

This project estimates the probability that a basketball shot is made using shot context, location, defensive pressure, timing, and player movement information. The workflow began with a naive make-rate baseline and progressively richer logistic models before moving to XGBoost to capture nonlinear relationships and feature interactions. The final model is evaluated on a held-out season and then applied to the testing set to generate one probability for every shot.

1	2	3	4	5
RAW SHOT DATA	FEATURE ENGINEERING	SEASON-BASED VALIDATION	XGBOOST MODEL	CALIBRATION + TEST PREDICTIONS

FEATURE ENGINEERING

Defender approach

The string-valued closest-defender approach field is split into four numeric snapshots: 1.00, 0.75, 0.50 and 0.25 seconds before the shot.

Defensive context

Contest status, number of contesters and closest-defender distance provide direct information about pressure on the shooter.

Shot context

Distance, court location, shot type, shot clock, dribbles, shooter speed and game state describe the situation in which the attempt occurred.

MODEL DEVELOPMENT

Baseline

Overall training make rate used as the same prediction for every validation shot.

Logistic regression

Established whether standard shot and defender variables carried useful linear signal.

Engineered logistic models

Added defender approach, defender-distance change and shooter ability to test additional signal.

Final XGBoost

Tree-based model selected to capture nonlinear effects and interactions while optimizing log-loss.

WHY XGBOOST? The engineered features moved the linear models only slightly, suggesting that remaining predictive value was more likely to appear through nonlinear relationships and interactions. A shooter-ability in the XGBoost experiment also showed signs of overfitting, so that feature was excluded from the final model.

MODEL RESULTS

Model	Validation Log-Loss
Naïve baseline	0.6899
Best logistic regression	0.6423
Final XGBoost (submitted model)	0.6289

MODEL VALIDATION & INTERPRETATION

WHAT DRIVES THE MODEL?

Shot distance is the dominant source of gain in the final model. Closest-defender distance, contest status and shot type also contribute meaningfully. The defender-approach snapshots remain in the model, but their smaller importance indicates that they refine predictions rather than replace the basic geometry and defensive context of the shot.

Interpretation: The model's behavior is basketball-plausible: where the shot is taken and how much defensive pressure is present matter most.

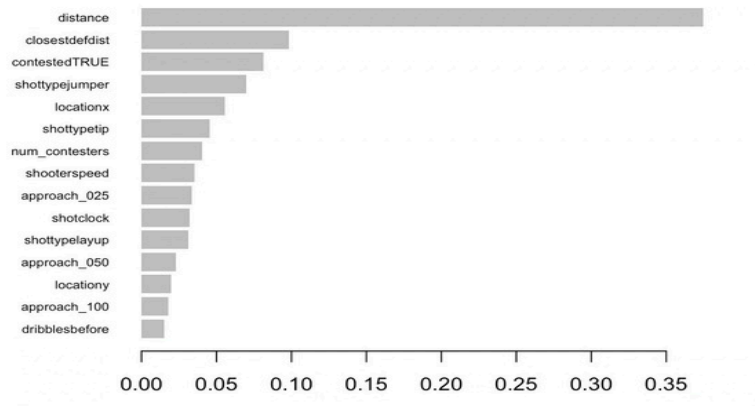
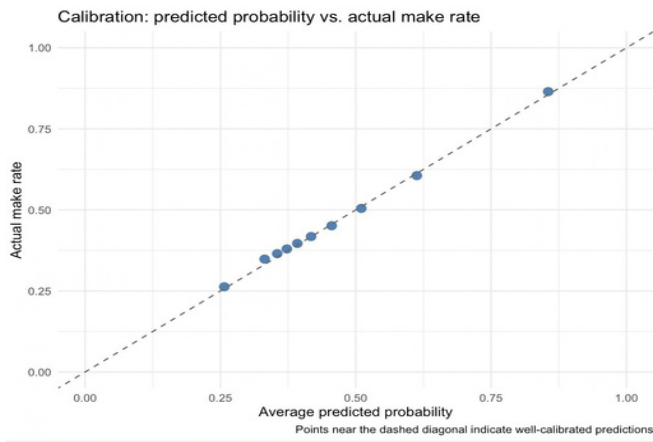
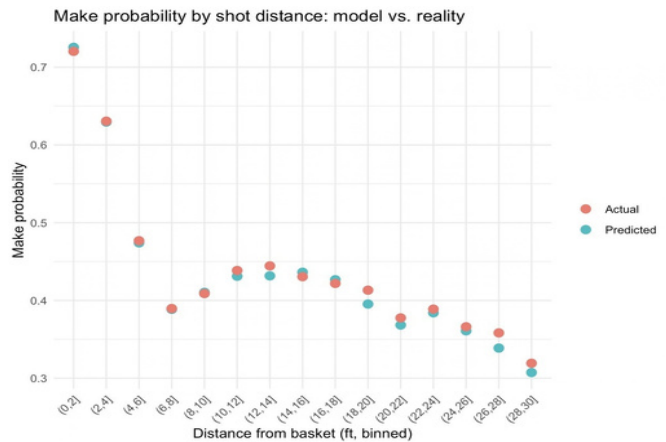


Figure 1. XGBoost feature importance measured by gain.



Calibration. Predicted probabilities closely track observed make rates across probability buckets.



Distance check. Predicted and actual make rates remain close across two-foot distance bins.

SEASON-BASED

Validation split

22 FEATURES

Final model matrix

0 MISSING

Values in final test matrix

0.6289

Final validation log-loss

CONCLUSIONS & NEXT STEPS

Strengths

The validation design mirrors the real task by training on one season and testing generalization on another. Calibration and distance-level checks show that the probabilities behave sensibly, not just that the model can rank shots.

Limitations

Player-specific ability was not retained in the final XGBoost model because the attempted encoding overfit. The model also relies on the supplied tracking/context variables and does not include richer player, lineup or possession history.

Future improvement

Use stronger shrinkage or out-of-fold target encoding for shooter ability, tune XGBoost with systematic cross-validation, and test additional interactions such as shot type by distance and defender movement by shot location.

Submission pipeline

The testing data is transformed with the same feature engineering and model-matrix structure as training. XGBoost then produces a make probability for every shot, preserving the original test-row order for submission.

Final takeaway: The strongest predictive signal came from shot geometry and defensive context, while XGBoost combined those signals into well-calibrated shot-level probabilities. One challenge was incorporating player-specific shooter ability: the attempted encoding showed signs of overfitting, so it was excluded from the final model. What impressed me most was how much shot distance dominated feature importance and how closely predicted probabilities matched actual make rates. The final model improved validation log-loss from 0.6899 for the naive baseline to 0.6289, showing a meaningful improvement while still generalizing well to the held-out season.